

Paper 2 - Set 01 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	B	Differentiation
2	C	Triangles, radians and sectors
3	F	Polynomials, factor and remainder theorems
4	C	The logic of arguments
5	E	Pythagoras, mensuration and 3D
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14	A	Identifying errors in proofs
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18	E	Exponentials and logarithms
19	A	Mathematical proof
20	F	Chains of reasoning

Worked solutions

Question 1 DIF-06 · Differentiation · spec MM6.1

The displacement s metres of a particle at time t seconds is $s = t^3 - 6t^2 + 9t$. At what times is the particle momentarily at rest?

Answer: B

$$\text{Velocity} = \frac{ds}{dt} = 3t^2 - 12t + 9 = 3(t - 1)(t - 3) = 0 \text{ at } t = 1 \text{ and } t = 3.$$

Full method - how and why

'At rest' means velocity zero, and velocity is the derivative of displacement.

1 Differentiate the displacement: $v = \frac{ds}{dt} = 3t^2 - 12t + 9$.

why: Velocity is the rate of change of displacement.

2 Factorise: $3(t^2 - 4t + 3) = 3(t - 1)(t - 3)$.

why: Take the 3 out first.

3 Set $v = 0$: $t = 1$ and $t = 3$.

why: 'Momentarily at rest' is exactly $v = 0$.

Where it goes wrong

- Setting $s = 0$ instead of $v = 0$, which gives $t = 0$ and $t = 3$ - an offered option.
- Forgetting one of the two roots.

Takeaway Displacement $\rightarrow$ velocity $\rightarrow$ acceleration by differentiating; 'at rest' means the velocity is zero, not the displacement.

Question 2 TRI-03 · Triangles, radians and sectors · spec MM4.2

A sector of a circle of radius 6 cm has area $15\pi \text{ cm}^2$. Find the arc length of the sector.

Answer: C

$$\frac{1}{2}r^2\theta = 18\theta = 15\pi \Rightarrow \theta = \frac{5\pi}{6}. \text{ Arc} = r\theta = 6 \cdot \frac{5\pi}{6} = 5\pi.$$

Full method - how and why

Sector area gives the angle; the angle gives the arc length. Both formulas need radians.

1 Use the sector area formula $A = \frac{1}{2}r^2\theta$: $\frac{1}{2}(36)\theta = 18\theta$.

why: $r = 6$, so $\frac{1}{2}r^2 = 18$. The formula is only valid with θ in radians.

2 Set equal to the given area: $18\theta = 15\pi$, so $\theta = \frac{5\pi}{6}$.

why: Divide: $\frac{15\pi}{18} = \frac{5\pi}{6}$.

3 Use the arc-length formula $s = r\theta = 6 \times \frac{5\pi}{6} = 5\pi \text{ cm}$.

why: The 6 cancels neatly - a sign the angle is right.

Where it goes wrong

- Quoting the angle $\frac{5\pi}{6}$ as the arc length - it is offered as a distractor.
- Using $A = r^2\theta$ without the $\frac{1}{2}$, which halves the angle and gives $\frac{5\pi}{2}$.

Takeaway $A = \frac{1}{2}r^2\theta$ and $s = r\theta$ (radians); find θ from one, then substitute into the other.

Question 3 POL-01 · Polynomials, factor and remainder theorems · spec MM1.6

When $x^3 - 4x^2 + kx + 6$ is divided by $(x - 3)$ the remainder is -6 . Find k .

Answer: F

Remainder Theorem: $27 - 36 + 3k + 6 = -6 \Rightarrow 3k - 3 = -6 \Rightarrow k = -1$.

Full method - how and why

A remainder is given, so use the Remainder Theorem: substitute the root of the divisor and set the result equal to the remainder.

- 1 The divisor $(x - 3)$ is zero at $x = 3$.

why: The Remainder Theorem says the remainder on dividing by $(x - a)$ is $f(a)$; you need the value of x that makes the divisor vanish.

- 2 Substitute: $f(3) = 27 - 36 + 3k + 6 = 3k - 3$.

why: Work the numerical terms out first ($27 - 36 + 6 = -3$) so only the k term is left to carry.

- 3 Set equal to the given remainder: $3k - 3 = -6$.

why: This is what the Remainder Theorem gives you - an equation, not an identity.

- 4 Solve: $3k = -3$, so $k = -1$.

why: Simple linear equation.

Where it goes wrong

- Substituting $x = -3$ (the sign of the root of $x - 3$ is $+3$), which gives a different value of k .
- Setting $f(3) = 0$ out of habit - that is the Factor Theorem, and only applies when the remainder is zero.

Takeaway Divisor $(x - a) \Rightarrow$ substitute $x = a$; the result is the remainder, which is zero only when $(x - a)$ is a factor.

Question 4 LGC-07 · The logic of arguments · spec Arg2

Let P : 'the quadrilateral $ABCD$ is a square' and Q : 'the quadrilateral $ABCD$ has four equal sides'. Which one of the following is true?

Answer: C

A square has four equal sides ($P \Rightarrow Q$, sufficient), but a rhombus that is not a square shows Q does not imply P (not necessary).

Full method - how and why

The same two-direction test, now in geometry. Look for a shape satisfying one condition but not the other.

- 1 Sufficiency: every square has four equal sides, so $P \Rightarrow Q$. **True.**

why: Straight from the definition of a square.

- 2 Necessity: does four equal sides force a square? A rhombus with angles 60° and 120° has four equal sides and is not a square.

why: A square also needs right angles; equal sides alone do not fix the angles.

- 3 So P is sufficient but not necessary for Q .

why: P is the stronger condition, so it implies the weaker one but not conversely.

Where it goes wrong

- Confusing which is the stronger condition: the more demanding property (square) is sufficient for the less demanding one (four equal sides).
- Failing to produce a concrete counterexample and guessing.

Takeaway A stronger condition is sufficient for a weaker one; the weaker is necessary for the stronger. Name a counterexample to confirm.

Question 5 MEN-11 · Pythagoras, mensuration and 3D · spec M5.16

Which of the following statements is/are true?

- I** Doubling the radius of a circle doubles its circumference.
- II** Doubling the radius of a sphere multiplies its volume by 8.
- III** Doubling the radius of a cylinder while halving its height leaves its volume unchanged.

Answer: E

$C = 2\pi r$ is linear in r (I). $V \propto r^3$ (II). $\pi(2r)^2 \cdot \frac{h}{2} = 2\pi r^2 h$: doubled, not unchanged (III false).

Full method - how and why

Three scaling claims. Each is checked by substituting the new dimensions into the formula.

- 1** I: $C = 2\pi r$ is linear in r , so doubling r doubles C . **True.**
why: Circumference scales with the first power of the radius.
- 2** II: $V = \frac{4}{3}\pi r^3$, so doubling r multiplies the volume by $2^3 = 8$. **True.**
why: Volume scales with the cube.
- 3** III: new volume $= \pi(2r)^2(\frac{h}{2}) = \pi(4r^2)(\frac{h}{2}) = 2\pi r^2 h$ - **double**, not unchanged. **False.**
why: The radius is squared, so doubling it quadruples that factor; halving the height only removes a factor of 2.
- 4** I and II only.
why: Match to the option list.

Where it goes wrong

- Assuming the doubling and halving in III cancel - the radius is squared, so they do not.
- Confusing area and volume scaling in II.

Takeaway Substitute the new dimensions into the formula; squared quantities change faster than linear ones.

Question 6 LGC-21 · The logic of arguments · spec Arg3

Consider the statement S : **for all real x , $x^2 - 6x + 10 > 0$** . Which of the following is true?

Answer: D

Completing the square shows the expression is at least 1 for every x .

Full method - how and why

A universal claim about a quadratic. Completing the square settles it, and the question also tests whether the **reason** given is sound.

- 1** Complete the square: $x^2 - 6x + 10 = (x - 3)^2 - 9 + 10 = (x - 3)^2 + 1$.
why: Halve the coefficient of x to get -3 , then correct by $-(-3)^2 = -9$.
- 2** $(x - 3)^2 \geq 0$, so the expression is $\geq 1 > 0$ for every real x . **True.**
why: A square is never negative; the minimum value is 1 at $x = 3$.
- 3** Check the false reasons: $x = 3$ gives $9 - 18 + 10 = 1$, not 0; and the discriminant $36 - 40 = -4 < 0$, so there are **no** real roots.
why: The claim that it has two real roots is simply wrong, and would in any case not prove positivity.

Where it goes wrong

- Assuming any quadratic takes negative values - one with a positive leading coefficient and negative discriminant never does.
- Mis-completing the square as $(x - 3)^2 + 10$ or $(x - 6)^2 + 1$.

Takeaway Completing the square proves positivity outright: $a(x - h)^2 + k$ with $a > 0, k > 0$ is always positive.

Question 7 X2-27 · Mathematical proof · spec Prf1

Consider the claim: **if n is a positive integer then $n^2 + n + 1$ is odd**. Which of the following is a valid proof?

Answer: A

A is a direct general argument. B only checks cases; C is irrelevant; D is correct for odd n but says nothing about even n ; E does not reach a contradiction.

Full method - how and why

Same 'which is a valid proof' pattern as PRF-01: general argument versus cases, irrelevance and incompleteness.

- 1 The valid proof: $n^2 + n = n(n + 1)$ is a product of consecutive integers, hence even; adding 1 gives an odd number.
why: General, covering every positive integer with no case split.
- 2 Checking $n = 1, 2, 3$ gives 3, 7, 13 - evidence, not proof.
why: Three cases out of infinitely many.
- 3 'It cannot be factorised, so it is odd' - factorisability has nothing to do with parity ($6 = 2 \times 3$ and $9 = 3 \times 3$ are respectively even and odd).
why: An irrelevant premise cannot support the conclusion.
- 4 'If n is odd then ... so $n^2 + n + 1$ is odd' - correct but only for odd n ; even n is untreated. Incomplete.
why: A case-based proof must cover every case.
- 5 'Suppose $n^2 + n + 1$ is even. Then n is even, contradiction' - n being even is not a contradiction; n can be even.
why: A contradiction proof must actually reach an impossibility.

Where it goes wrong

- Accepting the odd- n argument because every line in it is true - truth is not completeness.
- Accepting the contradiction attempt because it uses the right vocabulary.

Takeaway $n(n + 1)$ is always even, so $n^2 + n + 1$ is always odd. A proof must be general, relevant and complete.

Question 8 STA-05 · Statistics · spec M6.4

Which of the following statements about a scatter graph showing strong positive correlation between hours of revision and exam mark is/are valid?

- I Students who revised more tended to score higher.
- II Revising more causes a higher mark.
- III A line of best fit could be used to estimate the mark of a student who revised for a number of hours within the range of the data.

Answer: F

Correlation describes a tendency (I) and supports interpolation within the data range (III), but does not by itself establish causation (II).

Full method - how and why

A correlation-and-causation statement test. Decide what correlation does and does not licence.

- 1 I: strong positive correlation means higher revision went with higher marks. **True** - that is exactly what correlation describes.

why: 'Tended to' is a statement about the trend, not about every student.

- 2 II: correlation does not establish causation. **False.**

why: A hidden variable (motivation, prior ability) could drive both, so the data alone cannot prove cause.

- 3 III: using a line of best fit **within** the data range is interpolation. **True.**

why: Interpolation is legitimate; extrapolation beyond the data is what is unreliable.

- 4 So I and III only.

why: Match to the list.

Where it goes wrong

- Accepting II because the causal story sounds plausible - plausibility is not evidence.
- Rejecting III by confusing interpolation with extrapolation.

Takeaway Correlation licenses description and interpolation; it never licenses a causal claim on its own.

Question 9 PRF-04 · Mathematical proof · spec Prf1

A student wants to prove that **the square of any odd integer leaves remainder 1 when divided by 8**. Which of the following is the most appropriate first step?

Answer: E

A direct proof: $(2k + 1)^2 = 4k^2 + 4k + 1 = 4k(k + 1) + 1$, and $k(k + 1)$ is even, so the square is $8m + 1$. Checking cases is not a proof; assuming the claim is circular.

Full method - how and why

Choosing a proof strategy. For a 'for all odd integers' claim, the standard opening is an algebraic parametrisation.

1 Represent an arbitrary odd integer as $2k + 1$, k an integer.

why: This one expression covers **every** odd integer, which is what a universal claim needs.

2 (For context, the proof then runs: $(2k + 1)^2 = 4k^2 + 4k + 1 = 4k(k + 1) + 1$.)

why: Factorising $4k^2 + 4k$ as $4k(k + 1)$ exposes the consecutive pair.

3 $k(k + 1)$ is a product of consecutive integers, hence even, say $2m$, so the square is $8m + 1$.

why: That is remainder 1 on division by 8, as claimed.

Where it goes wrong

- Checking 1, 3, 5, 7, 9 - evidence, not proof.
- Writing the odd integer as $2k$, which is the form of an **even** integer.
- Assuming the claim is true - circular.
- Reaching for contradiction ('assume the remainder is not 1'): legal, but here a direct proof is simpler, and the question asks for the most appropriate step.

Takeaway Universal claims about odd/even numbers start by writing the general form: $2k$ or $2k + 1$.

Question 10 ERR-09 · Identifying errors in proofs · spec Err2

A student proves that the equation $x^2 + 2x + 3 = 0$ has no real solutions.

- I Complete the square: $x^2 + 2x + 3 = (x + 1)^2 + 2$.
- II $(x + 1)^2 \geq 0$ for all real x .
- III So $(x + 1)^2 + 2 \geq 2 > 0$ for all real x .
- IV Hence the equation has no real solutions.

Which of the following is correct?

Answer: A

Every step is valid.

Full method - how and why

Another valid proof. Completing the square and bounding below is a standard, correct technique.

- 1 Line I: $(x + 1)^2 = x^2 + 2x + 1$, so $x^2 + 2x + 3 = (x + 1)^2 + 2$ ✓.
why: Half the coefficient of x is 1; correct by $3 - 1 = 2$.
- 2 Line II: any real square is ≥ 0 ✓.
why: True for every real x .
- 3 Line III: adding 2 gives ≥ 2 , and $2 > 0$ ✓.
why: Adding a constant to both sides of an inequality is always valid.
- 4 Line IV: an expression that is always at least 2 can never equal 0, so there are no real roots ✓.
why: That is the definition of having no real solutions. (Check: discriminant $4 - 12 = -8 < 0$ ✓.)

Where it goes wrong

- Claiming $(x + 1)^2$ can be negative - not for real x .
- Claiming 'a positive expression can still equal zero' - it cannot; that is what positive means.
- Assuming there must be an error.

Takeaway Completing the square gives a rigorous lower bound; if the minimum is positive, there are no real roots.

Question 11 PRO-03 · Probability · spec M7.7

$P(A) = 0.6$, $P(B) = 0.5$ and $P(A \text{ and } B) = 0.3$. Which of the following statements is/are true?

- I A and B are independent.
- II A and B are mutually exclusive.
- III $P(A \text{ or } B) = 0.8$

Answer: F

$P(A)P(B) = 0.3 = P(A \text{ and } B)$, so independent (I). Not mutually exclusive since they can both happen (II false).
 $P(A \text{ or } B) = 0.6 + 0.5 - 0.3 = 0.8$ (III).

Full method - how and why

Three definitions to test: independence, mutual exclusivity and the addition rule.

- 1 I: independence means $P(A \cap B) = P(A)P(B)$. Here $0.6 \times 0.5 = 0.3$, which matches. **True.**

why: Independence is a numerical condition, not something you can eyeball.

- 2 II: mutually exclusive means $P(A \cap B) = 0$. It is $0.3 \neq 0$. **False.**

why: They overlap, so they can happen together.

- 3 III: $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6 + 0.5 - 0.3 = 0.8$. **True.**

why: Subtract the overlap or you count it twice.

- 4 So I and III only.

why: Match to the list.

Where it goes wrong

- Treating independent and mutually exclusive as the same thing - they are near-opposites: two events with non-zero probabilities cannot be both.
- Forgetting the $-P(A \cap B)$ and answering 1.1, which is impossible for a probability.

Takeaway Independent: $P(A \cap B) = P(A)P(B)$. Mutually exclusive: $P(A \cap B) = 0$. Check the numbers, not the words.

Question 12 FUN-12 · Functions and transformations · spec MM1.7

Let $f(x) = \sqrt{x-3}$ for $x \geq 3$ and $g(x) = x^2 + 3$. Which of the following statements is/are true?

- I $g(f(x)) = x$ for all $x \geq 3$
- II $f(g(x)) = x$ for all real x
- III $f(g(x)) = |x|$ for all real x

Answer: F

I: $g(f(x)) = (\sqrt{x-3})^2 + 3 = x$ for $x \geq 3$: true. $f(g(x)) = \sqrt{x^2 + 3} = |x|$, so II fails (e.g. $x = -1$) and III is true.

Full method - how and why

Two functions that almost invert each other. The asymmetry comes from $\sqrt{x^2} = |x|$ and from the restricted domain.

1 I: $g(f(x)) = (\sqrt{x-3})^2 + 3 = (x-3) + 3 = x$ for $x \geq 3$. **True.**

why: Squaring a square root recovers the inside, provided the inside is non-negative - guaranteed by the stated domain $x \geq 3$.

2 Compute the other order: $f(g(x)) = \sqrt{(x^2 + 3) - 3} = \sqrt{x^2} = |x|$.

why: Do the algebra once and read both remaining statements off it.

3 II claims this equals x for all real x : false at $x = -1$, where $|x| = 1 \neq -1$. **False.**

why: The square root returns the non-negative value, so it cannot reproduce a negative input.

4 III says it equals $|x|$, which is exactly what we found. **True.**

why: Same computation, correct conclusion.

5 I and III only.

why: Match to the options.

Where it goes wrong

- Assuming $f(g(x)) = x$ because f and g 'look like' inverses - inverses require matching domains and ranges.
- Writing $\sqrt{x^2} = x$.

Takeaway $f(g(x)) = x$ and $g(f(x)) = x$ are different statements; $\sqrt{x^2} = |x|$ is what breaks the symmetry here.

Question 13 ERR-11 · Identifying errors in proofs · spec Err1

Claim: $\int_{-1}^1 \frac{1}{x^2} dx = -2$.

- I An antiderivative of x^{-2} is $-x^{-1}$.
- II So the integral equals $\left[-\frac{1}{x}\right]_{-1}^1 = -1 - 1 = -2$.
- III Hence the value is -2 .

Which of the following is correct?

Answer: A

The integrand is positive everywhere it is defined, so a negative answer is a warning sign; the function is not defined on the whole interval and the integral does not exist in the ordinary sense.

Full method - how and why

The fundamental theorem of calculus needs the integrand to be defined (and continuous) throughout the interval.

- 1 Spot the warning sign: $\frac{1}{x^2} > 0$ wherever it is defined, yet the 'answer' is negative.
why: An integral of a positive function over an interval cannot be negative - the sign alone signals an error.
- 2 Line I is fine: $\frac{d}{dx}(-x^{-1}) = x^{-2}$ ✓.
why: Differentiate: $-(-1)x^{-2} = x^{-2}$.
- 3 Line II applies the fundamental theorem across $[-1, 1]$. But $\frac{1}{x^2}$ is undefined at $x = 0$, which lies **inside** the interval. **First error.**
why: The antiderivative is not continuous on the whole interval, so the theorem does not apply.
- 4 The arithmetic in line II is itself right ($-\frac{1}{1} - (-\frac{1}{-1}) = -1 - 1 = -2$); the fault is applying the theorem at all.
why: Correct arithmetic on an invalid application still gives nonsense.

Where it goes wrong

- Trying to 'fix' the sign to $+2$ - the integral does not exist in the ordinary sense, so no finite value is right.
- Blaming line I; the antiderivative is correct.
- Ignoring the discontinuity because the limits themselves are perfectly ordinary numbers.

Takeaway Before applying $\int_a^b f = F(b) - F(a)$, check f is defined on all of $[a, b]$; and a positive integrand can never give a negative integral.

Question 14 ERR-07 · Identifying errors in proofs · spec Err2

A student writes the following argument.

I $0.1 > 0.01$.

II Taking logarithms to base 10 of both sides: $-1 > -2$.

III Multiplying both sides by $\log_{10} \frac{1}{2}$, which is negative, reverses the inequality:
 $-\log_{10} \frac{1}{2} < -2\log_{10} \frac{1}{2}$.

Which of the following is correct?

Answer: A

Logarithms are increasing functions, so line II is valid; $\log_{10} \frac{1}{2} < 0$, so reversing the inequality in line III is correct.

Full method - how and why

Sometimes the argument is simply valid. Check each step against the rules rather than assuming a trap.

1 Line I: $0.1 > 0.01$ ✓.

why: $\frac{1}{10} > \frac{1}{100}$.

2 Line II: $\log_{10}$ is an **increasing** function, so it preserves the inequality: $\log_{10} 0.1 = -1$ and $\log_{10} 0.01 = -2$, and indeed $-1 > -2$ ✓.

why: Applying an increasing function to both sides keeps the direction; $\log_{10} 10^{-1} = -1$.

3 Line III: $\log_{10} \frac{1}{2} < 0$ because $\frac{1}{2} < 1$. Multiplying an inequality by a negative number reverses it, which is exactly what is written ✓.

why: $-1 > -2$ becomes $-1 \times c < -2 \times c$ for $c < 0$.

4 So all three lines are correct.

why: The question is testing whether you can certify a valid argument, not only spot faults.

Where it goes wrong

- Assuming logarithms reverse inequalities - they do not; log is increasing on its domain.
- Thinking $\log_{10} \frac{1}{2}$ is positive; logs of numbers below 1 are negative.
- Misreading $\log_{10} 0.1$ as 1 rather than -1 .
- Hunting for an error simply because the question implies one might exist.

Takeaway Increasing functions preserve inequalities; multiplying by a negative reverses them. $\log_b x < 0$ exactly when $0 < x < 1$ (for $b > 1$).

Question 15 BIN-09 · Binomial expansion · spec MM2.4

Which of the following statements is/are true?

- I** $\binom{n}{r} = \binom{n}{n-r}$ for all integers $0 \leq r \leq n$
II $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$ for all integers $0 \leq r < n$
III $\sum_{r=0}^n \binom{n}{r} = 2^n$ for all positive integers n

Answer: H

I is the symmetry of Pascal's triangle; II is Pascal's rule (each entry is the sum of the two above it); III is $(1+1)^n$ expanded. All true.

Full method - how and why

Three standard properties of binomial coefficients. Each has a one-line reason and a one-line check.

- I:** $\binom{n}{r} = \binom{n}{n-r}$ - choosing r objects is the same as choosing the $n-r$ to leave. **True.**
why: Visible as the left-right symmetry of each row of Pascal's triangle.
- II:** $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$ - Pascal's rule, i.e. each entry is the sum of the two above it. **True.**
why: Check with numbers: $\binom{3}{2} + \binom{3}{3} = 3 + 1 = 4 = \binom{4}{3}$ ✓.
- III:** $\sum_{r=0}^n \binom{n}{r} = 2^n$ - put $x = 1$ in $(1+x)^n$. **True.**
why: Equivalently, a set with n elements has 2^n subsets, counted by size.
- All three: 'I, II and III'.**
why: Match to the options.

Where it goes wrong

- Doubting Pascal's rule because of the index shift on the right - test it with small numbers.
- Thinking the sum in III is 2^{n+1} by miscounting the $n+1$ terms.

Takeaway Symmetry, Pascal's rule and row sum 2^n are the three facts about $\binom{n}{r}$ worth knowing cold.

Question 16 LGC-16 · The logic of arguments · spec Arg3

Which of the following statements is/are true about real numbers?

- I** For all x and all y , if $x < y$ then $x^2 < y^2$.
II For all x there exists y such that $x < y$ and $x^2 < y^2$.
III There exist x and y with $x < y$ and $x^2 > y^2$.

Answer: G

I fails for $x = -2$, $y = 1$. II: take $y = |x| + 1$. III: $x = -2$, $y = 1$ works.

Full method - how and why

Three quantified statements about squaring and ordering. Negative numbers decide all three.

- I:** $x < y \Rightarrow x^2 < y^2$ for all reals? Take $x = -2$, $y = 1$: $-2 < 1$ but $4 > 1$. **False.**
why: Squaring is only order-preserving for non-negative numbers.
- II:** for each x , is there y with $x < y$ and $x^2 < y^2$? Take $y = |x| + 1$: then $y > x$ and $y > |x|$ so $y^2 > x^2$. **True.**
why: Choosing y larger than $|x|$ guarantees both conditions at once - and y may depend on x .
- III:** do some $x < y$ have $x^2 > y^2$? The counterexample from I works: $x = -2$, $y = 1$. **True.**
why: Reuse the work - the same pair settles both.
- So II and III.**
why: Match to the list.

Where it goes wrong

- Accepting I by testing only positive numbers.
- Rejecting II by picking a bad y ; you get to choose y , so choose a helpful one.

Takeaway $x < y \Rightarrow x^2 < y^2$ needs $0 \leq x$; and to prove $\forall x \exists y$, construct y as a formula in x .

Question 17 TRG-05 · Trigonometric functions and equations · spec MM4.4

Which of the following statements about the graph of $y = \cos x$ is/are true?

- I It is symmetric about the y-axis.
- II It is a translation of the graph of $y = \sin x$ by $\frac{\pi}{2}$ in the negative x -direction.
- III It has period π .

Answer: E

$\cos(-x) = \cos x$ so I is true. $\sin(x + \frac{\pi}{2}) = \cos x$, so II is true. The period of $\cos$ is 2π , so III is false.

Full method - how and why

Three properties of the cosine graph. Each is a standard fact; the false one is about the period.

- 1 I: $\cos(-x) = \cos x$, so the graph is unchanged by reflection in the y-axis. **True.**
why: Cosine is an even function - its graph is symmetric about the y-axis, unlike sine.
- 2 II: $\sin(x + \frac{\pi}{2}) = \cos x$, and replacing x by $x + \frac{\pi}{2}$ shifts the graph $\frac{\pi}{2}$ to the left. **True.**
why: Check a point: $\sin$ at $x = \frac{\pi}{2}$ equals 1, and $\cos$ reaches 1 at $x = 0$ - shifted left by $\frac{\pi}{2}$ ✓.
- 3 III: the period of $\cos$ is 2π , not π . **False.**
why: $\cos(x + \pi) = -\cos x$, so a shift of π flips the graph rather than repeating it.
- 4 I and II only.
why: Match to the option list.

Where it goes wrong

- Thinking the period is π because $\cos$ returns to zero every π - the **whole** shape must repeat.
- Getting the direction of the shift in II backwards.

Takeaway $\cos$ is even with period 2π ; $\cos x = \sin(x + \frac{\pi}{2})$ is a shift to the left.

Question 18 X2-06 · Exponentials and logarithms · spec MM5.2

Which of the following is a counterexample to the claim 'for all positive x and y ,

$$\log_{10}(x + y) \leq \log_{10} x + \log_{10} y$$

Answer: E

$x = y = 1$: $\log 2 > 0 = \log 1 + \log 1$, so the inequality fails. For the others $x + y \leq xy$ and the claim holds (log is increasing).

Full method - how and why

To break an inequality claim, look for the case where the two sides are closest - here small values, where logs go to zero or negative.

- 1 Test $x = y = 1$: left side $\log_{10} 2 \approx 0.30$; right side $\log_{10} 1 + \log_{10} 1 = 0$.
why: $\log 1 = 0$ for any base, so the right side collapses.
- 2 $0.30 > 0$, so the claimed inequality $\leq$ fails. **Counterexample.**
why: Hypothesis satisfied (x, y positive), conclusion false.
- 3 Why the others work: $\log x + \log y = \log(xy)$, and log is increasing, so the claim is just $x + y \leq xy$. For $x = y = 2$: $4 \leq 4$ ✓; $x = y = 4$: $8 \leq 16$ ✓; $x = y = 5$: $10 \leq 25$ ✓; $x = 10, y = 100$: $110 \leq 1000$ ✓.
why: Converting the log inequality into an algebraic one makes all five cases quick to check.

Where it goes wrong

- Trying large values, where xy dwarfs $x + y$ and the claim comfortably holds.
- Forgetting $\log_{10} 1 = 0$ and miscomputing the right side.
- Thinking $\log(x + y) = \log x + \log y$ - that is the error the whole question is built on.

Takeaway $\log x + \log y = \log(xy)$, never $\log(x + y)$. Test inequality claims near the boundary, where the sides are closest.

Question 19 PRF-10 · Mathematical proof · spec Prf1

Consider the claim: **if x and y are real numbers with $x + y > 10$, then $x > 5$ or $y > 5$.** Which of the following proves it?

Answer: A

A is proof by contrapositive/contradiction. B assumes the conclusion, C is an example, D proves something false, E's conclusion ('both exceed 5') is wrong.

Full method - how and why

Proving 'if P then $(A \text{ or } B)$ '. Assuming both A and B fail and deriving a contradiction is the natural route.

1 Negate the conclusion: 'not $(x > 5 \text{ or } y > 5)$ ' means $x \leq 5$ **and** $y \leq 5$ (De Morgan).

why: Negating an 'or' gives an 'and' - and it gives you two usable facts instead of a disjunction.

2 Add the inequalities: $x + y \leq 10$.

why: Inequalities of the same direction may be added.

3 That contradicts the hypothesis $x + y > 10$, so the conclusion must hold.

why: Hypothesis true and conclusion false is impossible, which is exactly what proves the implication.

Where it goes wrong

- 'If $x > 5$ then the conclusion holds' - this assumes part of what is to be proved.
- Testing $x = y = 6$ - a single example proves nothing about all x, y .
- 'Suppose $x > 5$, then $y > 10 - x$, so $y > 5$ ' - the algebra is wrong and it proves a false statement (both exceeding 5).
- 'The average exceeds 5 so both exceed 5' - the average exceeding 5 only forces **at least one** to.

Takeaway To prove an 'or', negate it into an 'and' and look for a contradiction - De Morgan turns a weak conclusion into strong assumptions.

Question 20 RSN-08 · Chains of reasoning · spec Prf5

a , b and c are real numbers with $a < b < c$. Which of the following statements must be true?

I $a + b < 2c$

II $ab < bc$

III $\frac{a+c}{2} < c$

Answer: F

I: $a + b < c + c$. III: $\frac{a+c}{2} < \frac{c+c}{2} = c$. II fails when b is negative: $a = -3$, $b = -2$, $c = -1$ gives $ab = 6 > bc = 2$.

Full method - how and why

Inequality manipulation with 'must be true'. Two follow from replacing a term by a larger one; the third involves multiplying, which needs sign information.

1 I: from $a < c$ and $b < c$, add: $a + b < 2c$. **True.**

why: Adding two inequalities of the same direction is always valid.

2 III: $a < c$ so $a + c < 2c$, and dividing by 2 gives $\frac{a+c}{2} < c$. **True.**

why: Dividing by a positive constant preserves the direction. (The midpoint of two numbers lies below the larger.)

3 II: $ab < bc$ would follow from $a < c$ only if we could multiply by b - valid only for $b > 0$.

why: Multiplying an inequality by a negative number reverses it.

4 Counterexample: $a = -3$, $b = -2$, $c = -1$ satisfies $a < b < c$, but $ab = 6$ and $bc = 2$, so $ab > bc$. **False.**

why: All-negative values are the natural place to test a multiplication claim.

5 So I and III.

why: Match to the list.

Where it goes wrong

- Accepting II by 'multiplying $a < c$ by b ' without checking the sign of b .
- Testing only positive numbers, where II does hold.

Takeaway Adding inequalities is always safe; multiplying by a variable is not - you need its sign.